

Supplementary Information

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This file contains a glossary of official terms, Supplementary Notes 1–9, legends for Supplementary Tables 1–67 (provided as Supplementary_Tables.xlsx) and legends for Supplementary Data 1–3 (provided as separate Excel files).

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Glossary of official terms

These terms are used consistently in every figure, legend, table and data file. Retired wording is listed in the Glossary sheet of Supplementary_Tables.xlsx.

Study structure

Official term	Definition	Example in this study
Benchmark	A published collection of tasks with its own evaluator and endpoint. The three benchmarks are analysed separately and never pooled.	BixBench-Verified-50 (50 tasks); CompBioBench (100 tasks); IWC (10 tasks)
Platform-neutral benchmark	A benchmark whose tasks prescribe neither a workbench nor an analytical route.	BixBench-Verified-50; CompBioBench
Workflow-derived benchmark	A benchmark whose tasks are derived from curated, published Galaxy workflows and are therefore aligned with installed Galaxy tools by design.	IWC (Intergalactic Workflow Commission) benchmark
Task	One independently evaluated analytical problem within a benchmark, comprising the question or objective and the inputs needed for the analysis.	One BixBench question; one CompBioBench problem; one IWC workflow-derived task
Model configuration	The evaluated AI setup: the underlying model plus its agent harness and, where relevant, its reasoning setting. Always name the full configuration: the agent harness differs between configurations in this study, so "model" is never used as shorthand.	GPT-5.5; GPT-5.6 Sol; GPT-5.6 Luna; DeepSeek V4 Pro (Codex); DeepSeek V4 Pro (Claude Code, superseded; BixBench only)
Agent harness	The software that runs a model as an agent: it issues tool calls, executes commands and manages the conversation.	Codex (four model configurations); Claude Code (the superseded DeepSeek V4 Pro configuration)
Execution condition	The experimental arm defining how the agent may perform the computational analysis. The open-ended code condition is the reference condition and is shown first in every figure and table.	Galaxy condition; open-ended code condition
Galaxy condition	The arm in which the agent performs the analysis through the Galaxy workbench, using its programmatic interface and Galaxy analytical resources. Formal first-use name: Galaxy-mediated execution. The label records condition assignment; it does not certify that every operation ran inside Galaxy.	GPT-5.6 Sol solving a CompBioBench task through Galaxy
Open-ended code condition	The arm in which the agent freely writes and executes code and invokes command-line tools, packages and other resources, without being restricted to Galaxy. Formal first-use name: open-ended code execution.	Agent using Python, R, shell commands or installed packages
Run	One complete execution of one task by one model configuration under one execution condition in one replicate run. This is the atomic experimental observation.	GPT-5.6 Sol × bix-30-q3 × Galaxy condition × replicate run 2
Replicate run	One repetition of the same task × model configuration × execution condition. Replicate labels are not matched random seeds. CompBioBench replicates are final campaign selections and may include continuations.	Replicate run 1, 2 or 3
Replicate set	The replicate runs for one task × model configuration × execution condition (up to three runs). Classified by outcome repeatability as unanimous or split. Called the "cell" of the data table only once, in Methods.	The three Galaxy runs of GPT-5.6 Sol on bix-30-q3
Split replicate set	A replicate set whose replicate runs disagree in outcome: 1–2 of 3 scored correct (BixBench); 1–2 of 3 matching the consensus answer (CompBioBench, consensus proxy); or a within-set range of output agreement above 0.05 (IWC). A unanimous set has 3/3 or 0/3.	28 split Galaxy replicate sets versus 33 open-ended code sets in BixBench
Model–condition set	All runs for one model configuration under one execution condition within a benchmark; tasks and replicate runs vary.	All GPT-5.6 Sol Galaxy runs across the 50 BixBench tasks
Task–condition set	All runs for one task under one execution condition; model configurations and replicate runs vary.	All Galaxy runs for bix-30-q3
Condition-level summary	A result summarized across the evaluated model configurations, tasks and replicate runs for one execution condition within a benchmark.	Galaxy-condition accuracy on BixBench
Run population	The exact collection of runs included in a reported statistic, identifiable from benchmark, model configurations, conditions, tasks and replicate runs.	50 tasks × 4 model configurations × 3 replicate runs = 600 runs per condition

Outcomes and comparisons

Official term	Definition	Example in this study
Accuracy	The proportion of runs in a stated run population that the benchmark evaluator scored correct. Used only for BixBench, the one benchmark with binary evaluator verdicts. Runs that submitted no answer count as scored incorrect.	134 of 150 GPT-5.5 Galaxy runs scored correct on BixBench (89.3%)
Scored-correct run; scored-incorrect run	A run whose final answer the benchmark evaluator accepted (scored correct) or rejected (scored incorrect). "Rejected" describes the evaluator's action; the run itself is "scored incorrect".	246 scored-incorrect BixBench runs (111 Galaxy, 135 open-ended code)
Output agreement	A continuous measure (0–1) of agreement between a run's output and the reference output, used when the benchmark endpoint is continuous. Never converted to accuracy.	IWC output agreement for one run
Mean output agreement	The mean output agreement across a stated run population.	Mean IWC output agreement of 0.980 in the Galaxy condition (nine matched tasks)
Reported benchmark score	An aggregate score taken from archived answer vectors that cannot be reconstructed per run. It is neither accuracy nor output agreement.	CompBioBench, GPT-5.6 Sol Galaxy replicate run 1: 93/100 (labelled official)
Consensus proxy	For CompBioBench only: the most common answer across the 25 runs of a task, used to flag probable failures because per-run grades are not archived. Applied only to the 82 tasks where at least 20 of 25 runs agree, and always labelled as a proxy.	Reproduces reported benchmark scores with a mean absolute error of 2.6 of 100
Performance	Umbrella term for benchmark outcomes. Use it only when a statement holds for every endpoint involved; otherwise name the metric. Never use accuracy as the umbrella term.	"Performance was similar between execution conditions in all three benchmarks."
Score conflict	A run whose evaluator score differs from its run-record score, or where one of the two is missing.	17 of 240 IWC records
Condition difference	Galaxy condition minus open-ended code condition, holding the run population constant. The sign convention is always Galaxy – open-ended code.	+1.33 percentage points in accuracy on BixBench
Model difference	A comparison between model configurations under a specified execution condition and benchmark, always against a named reference model configuration.	GPT-5.6 Sol versus GPT-5.5 in the Galaxy condition: –0.67 percentage points
Task-level condition difference	A condition difference for one task, across a stated set of model configurations and replicate runs.	bix-45-q1: Galaxy 0/15 versus open-ended code 8/15 scored correct
Outcome repeatability	Consistency of outcomes (evaluator verdicts or output agreement) across the replicate runs of one replicate set.	Within-set range of IWC output agreement
Repeatability category	The categorical form of outcome repeatability for binary endpoints: 3/3, 1–2/3 or 0/3 replicate runs scored correct.	44 of 50 GPT-5.5 Galaxy replicate sets at 3/3 on BixBench
Answer consistency	Identical submitted answer text across replicate runs, with no implication of correctness. Used where evaluator verdicts are unavailable.	62 of 100 CompBioBench tasks with one distinct answer across all 12 Galaxy runs
Estimand	The specific quantity being summarized or compared. Used mainly in Methods.	Condition difference in accuracy

Analysis process and Galaxy records

Official term	Definition	Example in this study
Analysis trajectory	The sequence of analytical actions an agent takes during a run to reach, or attempt to reach, its final answer. It does not imply the answer was scored correct.	Inspect data → filter → run DESeq2 → inspect output → submit answer
Execution trace	The recorded evidence documenting an analysis trajectory: agent messages, commands, tool calls, Galaxy jobs and outputs.	Codex trace plus Galaxy history
Analytical operation	A scientifically meaningful computational action within an analysis trajectory.	Alignment, filtering, differential expression, enrichment
Tool-set fingerprint	An unordered set of tool identifiers (Galaxy) or command names (open-ended code) used in a run. It does not encode operation order, parameters or biological decisions.	Set of version-stripped Galaxy tool IDs used in one run
Tool-set similarity	Overlap between tool-set fingerprints across the replicate runs of a replicate set, measured as mean pairwise Jaccard similarity.	Mean Galaxy tool-set similarity of 0.616 (IWC) versus 0.134 (CompBioBench)
Declared route	The analytical method an agent states it used. It is not independently verified and is distinct from the tool-set fingerprint.	DESeq2 versus edgeR for IWC RNA-seq differential expression
Galaxy interface call	One call from the agent to Galaxy through the programmatic interface of the Galaxy condition (a Model Context Protocol server), such as a tool search, tool inspection or job submission.	69,812 Galaxy interface calls, of which 7,389 returned a failure status

Official term	Definition	Example in this study
Direct Galaxy API call	A shell command that calls Galaxy's web API or the BioBlend library directly, outside the Galaxy interface.	Copying the provided analysis history in 537 of 600 Codex BixBench Galaxy runs
Analysis history	The Galaxy record (history) of the datasets, jobs, tools and parameters produced during a run.	714 of 750 BixBench Galaxy runs have a detailed analysis-history snapshot
Galaxy analysis job	A Galaxy job other than a data-fetch (upload) job, counted once per server and job identifier.	23,080 Galaxy analysis jobs (5,042 BixBench; 16,686 CompBioBench; 1,352 IWC)
Installed Galaxy tool	A Tool Shed wrapper available on the Galaxy server. Subdivided into domain tools and utility tools.	Any wrapper in the server's tool panel
Domain tool	An installed Galaxy tool implementing a domain-specific analytical method.	fastp, DADA2, PepQuery, Bowtie2
Utility tool	An installed Galaxy tool for generic data handling.	Cut, Filter, xlsx2tsv, Datamash
User-defined tool (UDT)	A custom tool written by the agent and executed through Galaxy's interface.	Custom wrapper for PhyKIT treeness in bix-11-q1
Probe tool	A throwaway user-defined tool written only to find out why jobs fail, because Galaxy offers no trial run.	1,859 probe-tool calls in 517 of 808 CompBioBench runs that used user-defined tools
Parameter substitution	A difference between the parameter values an agent requested in a Galaxy tool-run call and the tool state Galaxy resolved from them (a value replaced, or a requested value dropped). It is either blocked before submission by the benchmark's check or executed.	4,352 tool-run calls, 916 executed (for example, Datamash max → count)
Operational error	A failed computational operation during a run. It does not by itself imply that the final answer was scored incorrect. Always reported through one of its two instrument-specific subtypes.	Galaxy job error; nonzero shell exit
Galaxy job error	A Galaxy analysis job that ends in error state. Not comparable with nonzero shell exits.	552 of 5,042 BixBench Galaxy analysis jobs
Nonzero shell exit	A shell command that returns a nonzero exit code. It can reflect probes or searches rather than analytical failure.	Failed package check or empty search result
Candidate recovery	An operational error followed by a later successful related action, not yet manually adjudicated. This is the default term.	93 candidate recoveries in BixBench
Verified recovery	A candidate recovery that manual review confirms: the agent overcame the failure while keeping the same analytical objective.	String columns cast to numeric before a repeated calculation (bedtools-chromhmm-q1)
Input-token usage	Input tokens consumed by a run or run population, according to archived usage records (cached input included; output and reasoning tokens not added).	Median of 1.19 million input tokens per GPT-5.5 Galaxy run on BixBench
Input-token ratio	Median Galaxy input-token usage divided by median open-ended code input-token usage, for the same task and model configuration.	Median ratio of 4.74 on BixBench

Adjudication

Official term	Definition	Example in this study
Primary cause; secondary cause	The main reason, and at most one contributing reason, for a scored-incorrect run (or an IWC run with output agreement below 0.5), assigned by trace-level adjudication from seven categories: task under-specified or reference ambiguous; statistical or reasoning error; evaluator rejected a correct answer; missing domain knowledge; platform or tool defect; no answer submitted; output format violated.	Galaxy platform or tool defect: primary in 4 and secondary in 18 of 111 scored-incorrect Galaxy runs
Adjudication confidence	How firmly the decisive step of a scored-incorrect run was established: high, moderate, mixed or unresolved (Supplementary Note 6).	159 high, 55 moderate, 20 mixed and 12 unresolved of 246 BixBench adjudications
Divergence mechanism	What separated a scored-incorrect replicate run from its scored-correct sibling(s) in a split replicate set. Distinct from its primary cause.	Hand-written method or software-version difference: 30 of 45 open-ended code runs versus 5 of 36 Galaxy runs
Verifier mode	The rule the BixBench evaluator uses to compare a numeric answer with the reference: tolerance (within a stated margin) or rounded-numeric (after rounding).	bix-43-q2: 5.831005... scored correct under the tolerance verifier and incorrect under the rounded-numeric verifier

Supplementary Note 1. Evidence archive and scope

The analysis is retrospective and read-only. It uses 4,240 runs: BixBench-Verified-50 (50 tasks; 1,500 runs: five model configurations × two execution conditions × three replicate runs), CompBioBench (100 tasks; 2,400 paired runs from four model configurations, plus 100 unpaired GPT-6 Astra runs in the open-ended code condition) and IWC (10 tasks; 240 runs, four model configurations). Of these, 4,228 retain an execution trace (Codex JSONL, or Claude Code stream-JSON for the superseded DeepSeek V4 Pro model configuration); the 12 missing traces are CompBioBench runs. Galaxy job records comprise 23,080 Galaxy analysis jobs (5,042 BixBench, 16,686 CompBioBench, 1,352 IWC), counted once per server and job identifier; data-fetch jobs are excluded. Detailed analysis-history snapshots exist for 714 of 750, 1,198 of 1,200 and 118 of 120 Galaxy-condition runs; they were taken after the runs completed.

Archive integrity was checked before analysis: 160 source-evidence hashes were verified, all 4,240 run identifiers are unique, and the archived accuracy, Galaxy job states and input-token medians were reproduced from the raw records. No score was regraded; the original evaluator verdict of every run is retained, and runs without an answer count as scored incorrect. No agent code was executed and no Galaxy server was contacted. BixBench reference values were read only from the evaluator records of completed runs.

The three benchmarks keep different endpoints: accuracy (BixBench-Verified-50, binary evaluator verdicts), reported benchmark scores without per-run grades (CompBioBench) and continuous output agreement with workflow reference outputs (IWC). They are analysed separately and never pooled. Extended Data Fig. 1a summarizes the archive and Extended Data Fig. 1b the audit pipeline.

Supplementary Note 2. Execution-trace extraction, call classes and direct Galaxy API calls

Every execution trace was parsed. For each Galaxy interface call we recovered the tool name, arguments, structured status, error text, failure summary, validation errors and parameter provenance. Calls were grouped as *tool search and inspection* (search_galaxy_tools, inspect_galaxy_tool), *analysis-history inspection* (history and archive inspection), *tool runs and jobs* (run_galaxy_tool_and_wait, run_galaxy_udt_and_wait, job waits) and shell commands. Input-token usage is recorded per run and cannot be attributed to individual calls, so the number of characters each call returned to the agent is used as a proxy for context consumed (Fig. 5b).

A shell command in a Galaxy-condition run is a direct Galaxy API call when it uses the BioBlend library or calls the Galaxy histories, datasets, jobs or tools API. Commands referencing BioBlend or any /api/ path were matched against five operations: copying the provided analysis history (copy_history, /copy), downloading datasets (download_dataset, /display, /download), reading tool parameter descriptions (show_tool, /build), checking job status (show_job, /jobs/{id}, wait_for_job) and submitting jobs directly (run_tool, or a POST to /api/tools through requests, curl or urllib). Direct Galaxy API calls, parameter substitution and user-defined-tool statistics use Codex execution traces (1,908 Galaxy-condition runs: 600 BixBench, 1,188 CompBioBench, 120 IWC), because the Claude Code agent harness records tool results in a different structure. A provided analysis history exists, and copying it is required by the harness policy, only in BixBench.

Supplementary Note 3. Causes of failed Galaxy interface calls

Of 69,812 Galaxy interface calls, 7,389 returned a failure status. Each failed call was assigned one cause by applying ordered regular-expression rules to its error text and status (first match wins; rules in Supplementary Table 34). Causes A1–A8 occurred before any job was created (3,454 calls); B1–B5 are failures of a created job, that is Galaxy job errors (3,677); 258 calls matched no rule and are reported as unclassified. Rule order resolves overlaps; for example, a missing analysis history is assigned A1 even when the message also contains a validation keyword. Each cause maps onto an interface change (Extended Data Fig. 3d).

Class	Cause	Definition
A1	Tool description needs an analysis history	Tool inspection returned "History unavailable" because no analysis history was specified.
A2	Tool identifier not found	A tool identifier that does not resolve on the server (usually guessed or truncated).
A3	Conditional option structure	Nested conditional or repeat parameters submitted with a key structure the server rejects.
A4	Parameter value or data type	A value, option, format or collection type rejected by tool-state validation.
A5	Dataset or analysis-history identifier	Wrong, foreign, truncated or undecodable dataset or analysis-history identifiers, or analysis histories not owned by the user.
A6	User-defined tool definition	User-defined tool definition rejected at creation (schema, pattern, discriminator or extra fields).
A7	Upload or file type	Unknown file extension or failed upload.

Class	Cause	Definition
A8	Server, connection or rate limit	Web error pages, uncaught server exceptions, closed connections, time-outs and rate limits.
B1	User-defined tool: software missing	Package or executable absent from the container chosen for a user-defined tool.
B2	Job failed, no error message	A Galaxy job error with no error message returned to the agent.
B3	Job failed with error message	A Galaxy job error that returned an error message or traceback.
B4	File format, compression or index	Input not parseable in the declared format, compression mismatch or missing index.
B5	Memory or compute limit	Memory exhaustion or a killed process.
Z	Unclassified	Failure text matched no rule; retained for completeness.

Supplementary Note 4. Parameter substitution

The Galaxy interface compares the parameter values an agent requested with the tool state Galaxy resolved from them, and returns the difference as parameter provenance. A parameter substitution is reported with one of two statuses: *validation_parameter_mismatch*, when the benchmark's check blocked submission, and *parameter_mismatch*, when the job executed. A value is *replaced* when a requested value was changed (for example *max* → *count*) and *dropped* when a requested parameter is absent from the resolved state, typically because it was placed under a conditional option or repeat that Galaxy did not select. Parameter substitution occurred in 4,352 tool-run calls: 3,436 blocked and 916 executed (Fig. 3e; Supplementary Table 37). Supplementary Data 3 lists the first replaced value of each call with a replaced value (2,085 calls). Whether a substitution changed a scientific result was established only for the adjudicated runs in the failure ledger.

Supplementary Note 5. User-defined tool reliability

Every run *_galaxy_udt_and_wait* call returns a structured status (ok, failed, *udt_creation_failed* or none). For each Galaxy job error, the failure record gives the phase (before execution or during command rendering, versus command runtime) and whether an error message was returned. A Galaxy job error with no error message was followed to the next tool-run call of the same run, which was classed as an identical request when its tool inputs or user-defined-tool definition were byte-identical after key sorting, and otherwise as changed. Probe tools were identified by identifiers containing probe, preflight, render, smoke, diagnos(tic) or sanity. Of the calls that followed a Galaxy job error with no error message, 784 resubmitted an identical request and 615 of those failed again (Extended Data Fig. 3a; Supplementary Table 28).

Supplementary Note 6. Failure adjudication protocol

Every scored-incorrect BixBench-Verified-50 run (246: 135 open-ended code condition, 111 Galaxy condition) and every IWC run with output agreement below 0.5 (8) was read call by call. For each, we identified the decision point: the step at which the evidence the agent acted on diverged from what the reference required, and whether that evidence was sufficient. Scored-incorrect runs were contrasted with scored-correct runs of the same task, using the same model configuration where one succeeded. Final answers were checked against independent evidence where possible (for example, reconstructing a statistic from the retained inputs, recomputing Benjamini–Hochberg adjustments, or comparing assembled contigs with the public *Agrius convolvuli* mitochondrial genome OZ203683.1 by canonical 31-base sequences) rather than against the agent's own conclusion.

Each run received one primary cause and at most one secondary cause. Galaxy is counted as *implicated* when a platform or tool defect is the primary or secondary cause: 0 of 135 scored-incorrect open-ended code condition runs and 22 of 111 scored-incorrect Galaxy-condition runs.

Cause	Definition	Open-ended code condition, primary / secondary (n = 135)	Galaxy condition, primary / secondary (n = 111)
Task under-specified or reference ambiguous	The question is under-specified, or the reference depends on a choice the question does not state	64 / 26	58 / 12
Statistical or reasoning error	Weak evidence relied on, thresholds or invariants ignored, or a wrong denominator	39 / 63	26 / 52
Evaluator rejected a correct answer	The evaluator rejected an equivalent answer, or the run has a score conflict	18 / 1	15 / 0
Missing domain knowledge	A domain-knowledge gap	12 / 6	3 / 3
Platform or tool defect	A defect of Galaxy, the Galaxy interface or its tools (for open-ended code condition runs: local software)	0 / 0	4 / 18
No answer submitted	No answer written (turn ended or budget exhausted)	1 / 0	5 / 0
Output format violated	Output-format or identifier violation	1 / 0	0 / 0

Adjudication confidence was assigned per run: **high** (159), the decisive step is visible in the execution trace and confirmed by reproduction, independent reference comparison or contrast with a scored-correct sibling; **moderate** (55), the decisive step is identified but only partly verified; **mixed** (20), several contributing causes are present and the primary assignment is a judgement; **unresolved** (12), no decisive step could be isolated and the cause is the best-supported explanation. Mixed and unresolved assignments (32 of 246) are not proofs. The failure ledger (Supplementary Table 14) records the decision point for every run. For IWC, the two Galaxy-condition host-removal runs with a score conflict (evaluator 0.273, run record 0.9999 and 1.0) were re-examined: the submitted BWA-MEM output kept 20,899 read pairs, matching the BWA-route reference (20,896) rather than the Bowtie2-route reference (72,867) against which it was scored; the IWC archive itself does not adjudicate these score conflicts (Extended Data Fig. 2d).

Supplementary Note 7. CompBioBench consensus proxy

CompBioBench archives 22 reported benchmark scores but no per-run grades or references, so per-task correctness cannot be observed. Answers were normalized (lower case, whitespace removed) and the most common answer across the 25 runs of each task was taken as the consensus answer. For the 22 reported benchmark scores with retained answers, the number of answers matching the consensus reproduces the reported score with a mean absolute error of 2.6 of 100 and a mean bias of +2.0 (Extended Data Fig. 1c). On the 82 tasks where at least 20 of 25 runs agree, a deviating answer is treated as a *probable* failure; this yields 40 open-ended code condition and 33 Galaxy-condition deviations (GPT-6 Astra excluded). The 18 contested tasks are not used. Re-fitting individual answers to the reported scores was rejected because it over-fits (22 constraints, 100 unknowns). The consensus proxy is labelled as such wherever it appears and supports no per-task accuracy claim.

Supplementary Note 8. Outcome repeatability: split replicate sets and divergence mechanisms

A replicate set is the replicate runs of one task × model configuration × execution condition (the "cell" of the data table). A replicate set is *split* when its replicate runs disagree in outcome: 1 or 2 of 3 BixBench-Verified-50 replicate runs scored correct (repeatability category 1–2/3); 1 or 2 of 3 CompBioBench replicate runs match the consensus answer (82 strong-consensus tasks); or the within-set range of IWC output agreement exceeds 0.05. For every scored-incorrect replicate run in a split BixBench-Verified-50 replicate set, the execution trace was compared with those of the scored-correct sibling(s) to identify the divergence mechanism: **Galaxy interface trap**, an optional interface path taken by this replicate run only produced a silent default, a different output semantics or an undispached job; **different software version installed**; **domain convention or definition applied differently**; **hand-written method instead of the library method** (a script, or a user-defined tool in the Galaxy condition); **error in the final step** (sorting, counting, units) after an otherwise correct analysis; **no answer submitted**; and **answer retrieved from benchmark source files**, when the only scored-correct replicate run retrieved benchmark source data. A divergence mechanism describes what differed between replicate runs and is distinct from the primary cause (Note 6). Replicate labels are not matched random seeds, so outcome repeatability means run-to-run agreement under the same prompt and model configuration.

Supplementary Note 9. Statistical analysis and limitations

Unless stated otherwise, intervals are exploratory 95% percentile cluster-bootstrap intervals with 20,000 resamples (seed 20260922; NumPy linear quantiles; statistic-specific SHA-256 streams). Resampling units are source capsules for BixBench (up to 33), tasks for CompBioBench (up to 100) and tasks for IWC; model configurations and replicate runs are retained within each cluster, and cross-benchmark draws are independent and stratified by benchmark. Intervals assume independent clusters, which is not established for shared biological inputs, and are pointwise rather than multiplicity-adjusted. No confirmatory significance, equivalence, non-inferiority or causal claim is made; an interval spanning zero (differences) or one (ratios) is inconclusive. The input-token ratio divides the median input-token usage of the three Galaxy-condition runs by the median of the three open-ended code condition runs for the same task and model configuration (all six values required), and is summarized by its median. Spearman correlations between prompt length or recorded workload and operational errors are across tasks, use the three shared GPT model configurations (nine observed runs required per task and execution condition) and 5,000 bootstrap resamples with ties re-ranked within each resample.

Limitations. Benchmarks differ in task selection, prompts, execution budgets, exposed interfaces and endpoints, so cross-benchmark contrasts are descriptive. The IWC estimates rest on nine or ten task clusters. The superseded Claude Code model configuration ran BixBench-Verified-50 only. Analysis histories were snapshotted after the runs. Adjudication was performed by execution-trace review, and 32 of 246 assignments are mixed or unresolved.

Candidate recoveries were not adjudicated, and the open-ended code condition has no equivalent job-level record. Nonzero shell exits are an operational marker, not a scientific failure rate. Human review time, reconstruction accuracy and monetary cost were not measured.

Supplementary Table legends

All tables are provided in Supplementary_Tables.xlsx, one sheet per table, with an index sheet mapping each table to its archive identifier and first citation, and a glossary sheet.

Supplementary Table 1 | What can be combined fairly? The historical CompBioBench source composes final vectors from multiple campaigns, unlike a prospectively fixed independent replicate design. IWC has unbalanced 6 h and 12 h budgets and execution condition-specific prompts. Prompt additions, harnesses, tools, model verification, domain mix and campaign selection remain confounders. No cross-benchmark pooled accuracy denominator is constructed. Archive table X1.

Supplementary Table 2 | Original-evaluator accuracy and paired execution condition difference. Original binary evaluator scores are retained, including six scored missing answers (five Galaxy, one open-ended code). No regrading was performed. The superseded Claude Code model configuration contributes 16 of the 24 extra scored correct Galaxy runs; excluding it leaves eight. The original 100,000-resample audit interval was -1.41 to +7.91 percentage points; the fresh calculation above uses the common settings below. Intervals as in Supplementary Note 9. Archive table B1.

Supplementary Table 3 | Task coverage versus within-model configuration advantage. Task coverage pools five model configurations and three replicate runs (15 runs/task/execution condition), whereas the last column compares three-replicate counts within each of 250 matched task/model configuration pairs. There are 198 tied pairs, and two tasks with no scored correct answer in either execution condition. These endpoints do not describe the reliability of a single run. Archive table B10.

Supplementary Table 4 | Tasks with evaluator accuracy in only one condition. Exact rule: at least one scored correct replicate across the five model configurations in one condition and zero across all 15 runs in the other. Galaxy-only: zero tasks. Open-ended-code-only: one task, bix-45-q1. This is a task-level set comparison, not a test of superiority. Relative composition variability is the alignment statistic requested in the task. Model-specific counts are shown next. Archive table B11.

Supplementary Table 5 | Source-reported reported benchmark scores, with provenance status. O = archive-labelled official, not an independently verified leaderboard receipt; P = predicted. 15 vectors are O and 10 P; 13 advertised hashes match retained bytes, nine mismatch and three vectors are absent. All parsed answers match for the 22 available vectors. Sol Galaxy r1/r3 were previously 92/100 P; dated metadata changes them to 93/100 O and 91/100 O. Neither version is silently discarded. No pooled accuracy, uncertainty interval, or task-level all/some/none-correct inference is computed from this mixture. Source: aggregate audit, score_vectors and score_conflicts. Archive table C1.

Supplementary Table 6 | CompBioBench consensus proxy and its validation against reported benchmark scores. CompBioBench keeps reported benchmark scores but no per-run grades. For each task the most common normalized answer across its 25 runs is the consensus answer; tasks where at least 20 of 25 runs agree have a strong consensus. The proxy is validated by counting, for each reported benchmark score with retained answers, how many of its 100 answers match the consensus (a). Deviations are probable, not adjudicated, failures. Source: CompBio/compBio_overview_audit.json and archived answers.

Supplementary Table 7 | Matched output agreement on the common nine IWC tasks. Output agreement is the saved iwc-final-answer-v2.2 reference_accuracy. Its definition and tolerance are task-specific (Supplementary Table 9), so a mean is a task-weighted summary, not biological accuracy. Every column uses the same nine tasks: 27 runs per execution condition/model configuration, 108 per execution condition in All four. Host removal is excluded from both execution conditions for all models because of null and conflicted route scores (Supplementary Table 13). This is a post hoc sensitivity population, not a prespecified primary endpoint; all-ten-task summaries remain in Supplementary Table 9 and Supplementary Table 55. Three unsupported-route scores remain unknown, never zero. Differences average replicate runs within task, then weight tasks equally. For All four, it averages the 36 task/model configuration differences. Intervals resample the nine tasks, 20,000 times, seed 20260922. The IWC report's 10,000-resample intervals (seed 20260923) differ slightly; the point estimates are identical and checked by the generator. With nine clusters, percentile coverage is approximate. No equivalence margin was prespecified. Archive table I1.

Supplementary Table 8 | Continuous replicate dispersion on the same nine tasks. One replicate set is one task, model configuration and execution condition with three numeric replicate runs; all columns use the Supplementary Table 7 population. Within-set range is maximum minus minimum across the three scores, summarized over nine replicate sets. These continuous summaries introduce no pass threshold. Small dispersion can accompany consistently low output agreement and is not correctness. With only three runs per replicate set, this is descriptive repeatability, not a stable variance estimate. Host-removal exclusions are retained in Supplementary Table 13; replicate labels are not matched seeds. Archive table I2.

Supplementary Table 9 | Task-level output agreement, zero-scored runs and Galaxy job errors. All ten tasks are shown because IWC has too few tasks for a selection rule. Each execution condition has 12 runs per task (four model configurations x three replicate runs). Minimums expose low-scoring runs that means can conceal; no distributional modality is assumed. Host removal has different model composition (12 versus nine numeric runs) and is not a matched contrast. Jobs are deduplicated within task and include legacy upload. Galaxy job errors coexist with near-ceiling output agreement: short-read QC has 22 Galaxy job errors yet a minimum Galaxy output agreement of 0.996. Score conflicts are records whose evaluator and run-record scores differ by more than 1e-9, or where one is null. Definitions come from retained evaluator metric records and are abbreviated here. Archive table I3.

Supplementary Table 10 | Does the same tool-set fingerprint instrument reproduce its tool-set similarity across benchmarks? Descriptive population: three shared GPT labels; all three run fingerprints observed and nonempty. This holds the extraction rule and included supplied labels fixed; eligible model configuration proportions can still differ. IWC's higher Galaxy tool-set agreement fits workflow-defined tool chains, but rests on ten tasks. It does not control biological task complexity, command-vocabulary coverage, custom-wrapper identity or campaign selection. Lower tool-set similarity describes more variable recorded toolsets/command indicators, not worse science. Instrument values must not be directly compared across the two rows. Archive table X3.

Supplementary Table 11 | Task origin, recorded helper exposure and detected tool use. All benchmark-specific paired model configurations. Requests, linkage and success use the Supplementary Table 24 rules. BixBench and CompBioBench exposure lists were not archived; observed requests show that the helper was available in at least those runs. Request rates and listed-run linkage counts have

different denominators. A linkage count divided by all listed runs is a detected lower bound, not an estimate that treats missing transcripts/histories as zero use. IWC has no named-helper detections and its recorded lists omit that helper; neither observation establishes absence of alternative custom-code interfaces. Tool Shed jobs outside the utility codebook (Supplementary Table 27) form a larger observed job share in IWC, but this classifier does not prove native completion of every task. Task origin and helper exposure are not experimentally separated. Local shell and interactive computation remain possible; Galaxy-only execution is uncertified in every benchmark. Archive table X19.

Supplementary Table 12 | How much of the execution condition difference rests on a few runs? Four decimals are kept because several differences are about $1e-3$. Sign stable means all nine leave-one-task-out estimates share the full estimate's sign. The zero-run column drops every task with a zero-scored run for that model configuration in either execution condition. It is an outcome-selected influence diagnostic, not a mechanism test or an improved performance estimate; it never replaces the full estimate. The last column adds host removal wherever all six runs are numeric. This drops the unmatched GPT-5.5 replicate set, and Luna then includes the two conflicted Galaxy records (Supplementary Table 13). The All four available-pair row weights 39 task/model configuration pairs equally, so host removal has three model configurations and other tasks four; it is not equal task weighting over ten tasks. The two larger positive differences shrink when tasks containing zero-scored runs are removed. Across model configurations, the remaining differences range from 0.0010 to 0.0124. Archive table I4.

Supplementary Table 13 | Score provenance that changes an IWC conclusion. This table plays the role of Supplementary Table 5: score lineage is part of the result. The conflict tolerance is an absolute difference above $1e-9$, or numeric versus null. Source: IWC scientific audit, `runs[*].score`, `run_record_acc` and `score_conflict_gt_1e_9`. Archive table I6.

Supplementary Table 14 | Failure ledger: every scored-incorrect BixBench-Verified-50 run and every IWC run with output agreement below 0.5. One row per run: 246 scored-incorrect BixBench-Verified-50 runs (135 open-ended code condition, 111 Galaxy condition) and 8 IWC runs with output agreement below 0.5. Primary and secondary causes: task under-specified or reference ambiguous (the reference depends on a choice the question does not state); statistical or reasoning error (weak evidence relied on, thresholds or invariants ignored, wrong denominator); evaluator rejected a correct answer (or the run has a score conflict); missing domain knowledge; platform or tool defect (Galaxy, the Galaxy interface or its tools; for open-ended code condition runs, local software); no answer submitted; output format violated. Adjudication confidence is defined in Supplementary Note 6.

Supplementary Table 15 | Model-configuration dependence of the exclusive open-ended code success. Single-task description: all 15 Galaxy answers were scored incorrect, although all 45 retained Galaxy analysis jobs had state ok. An condition-wide missing execution capability is therefore not established. No benchmark-wide interval is estimated from one task. Archive table B12.

Supplementary Table 16 | Why bix-45-q1 differs: a version contrast retained within one original run. These p-values are outputs of the biomedical task, not tests of an execution condition effect. The recorded checks use rounded per-alignment values and two-sided asymptotic/automatic Mann-Whitney tests with continuity correction. The agent attributed the change to version-dependent gap/ambiguous-residue handling; the archived numerical contrast supports a version explanation for this run. It does not prove the intended reference release, adjudicate biological validity, or explain every scored incorrect run. Source: task evidence, `open_ended_code_codex_gpt_5_6_sol_r1`, source lines 24, 33 and 45; trace messages and hashes are in `results.deep.case_sources`. No calculation was replayed. The within-run version contrast narrows the possible explanation beyond a generic input-assembly difference. Archive table B13.

Supplementary Table 17 | Evaluator-mode composition and accuracy. Modes are observational and task/model configuration-dependent; these are not randomized verifier comparisons. The source audit reports byte-identical answers with different scores under different modes. Rounded-numeric scoring occurs only in the two DeepSeek model configurations. These evaluator effects limit interpreting rejection as a demonstrated analytical mistake. Archive table B8.

Supplementary Table 18 | Completion is not correctness: every run scoring below 0.5. All 240 harness records report complete; completion alone does not establish final-artifact validity. The <0.5 selection is an auditor screening rule for inspection, not a benchmark failure threshold. Five of the eight runs are open-ended code runs. The only zero-scored Galaxy run is mitogenome assembly GPT-5.5 r1, and the same model configuration and replicate also scored zero in open-ended code. The other two Galaxy rows are score-conflicted host-removal records, not established scientific failures; final adjudicated values are unknown. Reasons summarize saved evaluator fields where present; casing is documented in the IWC report. No answer was repaired or regraded. Archive table I5.

Supplementary Table 19 | Biological domain, answer convergence and operational burden. Domains come from task metadata; these are not difficulty strata. All four paired model configurations are included. Single-answer counts use 12 submitted answers per task/execution condition. Error denominators include only detailed Galaxy runs. Ratios require all six usage records; small domains are descriptive case groups, not evidence of domain superiority. Archive table C4.

Supplementary Table 20 | Workbench capabilities visibly exercised by agents. All Galaxy model configurations with retrieved transcripts; nonexclusive run-level counts of named calls. A completed helper call does not by itself prove job success or scientific validity. History creation/copying is not universally measured by these named helpers; it is documented in selected traces and must not be inferred for every run from a linked history. These counts support feature use, not improved human readability. IWC user-defined-tool detections are zero and that helper is absent from recorded lists; this association does not establish why custom code was or was not used. Archive table X14.

Supplementary Table 21 | Most frequently recorded Galaxy operations. Top six tool IDs by job count (ties sorted by full ID), among 5,042 deduplicated Galaxy analysis jobs. Spreadsheet conversion, column selection and filtering accompany KEGG over-representation and PhyKIT metrics. KEGG is the Kyoto Encyclopedia of Genes and Genomes. Counts include preparation and inherited/late history state; operation frequency is not success attribution. Full IDs and event references are in the analysis JSON. Archive table B9.

Supplementary Table 22 | Most frequently recorded Galaxy operations. Top six tool IDs by job count among 16,686 deduplicated Galaxy analysis jobs. AnnData inspection and bedtools intersection appear alongside filtering, column selection, Datamash and legacy upload. This is evidence of operation availability/use, not correctness, independent analyses or exclusive use of installed domain tools. Full IDs and event references are in the analysis JSON. Archive table C8.

Supplementary Table 23 | Most frequently recorded Galaxy operations. Top six tool IDs by job count among 1,352 deduplicated Galaxy analysis jobs. Five of the six are domain workflow wrappers (read trimming, amplicon denoising, peptide verification, SRA metadata search), versus two of six in Supplementary Table 21 and Supplementary Table 22, where table utilities dominate. This fits tasks derived from Galaxy workflows. The top operation is still a text utility (awk). Counts include inherited or later history state; frequency is not success attribution. Archive table I10.

Supplementary Table 24 | User-defined tools: explicit requests, tasks reached and linked execution. An explicit request is a `run_galaxy_udt_and_wait` event. Linked execution requires a declared `GalaxyUserTool` identifier matching a retained Galaxy job tool ID in the same run; success requires job state ok. Matching IDs supports linkage but does not establish exact chronology or authorship for inherited jobs. Requests alone do not prove submission or execution; unobserved requests are unknown. Four YAML-string representations are not parsed and remain linkage gaps. Jobs invoked through other interfaces can be missed; these are lower-bound detections, not an exhaustive custom-tool inventory. The proposed 30-40% claim depends on its denominator: task coverage, run use and confirmed execution are not interchangeable. IWC has zero detected named-helper requests and the helper is absent from its 120 recorded lists; alternative interfaces and unparsed/missing events are not excluded (Supplementary Table 11). Official user-defined tool documentation (<https://galaxyproject.org/tools/user-defined-tools/>) describes this capability and recommends existing published tools when suitable. Archive table X12.

Supplementary Table 25 | Are user-defined tool choices stable across repeat executions? One replicate set is one task/model configuration; all three transcripts must be present. The three usage categories partition replicate sets. Same prompt means identical archived prompt-file hashes; the stricter subset additionally requires a single verified runtime ID and reasoning setting. Variation is compatible with stochastic or context-dependent choice, not proof of randomness: input provenance, tool catalogs, prior state, campaign selection and services were not controlled. Missing runtime metadata is not treated as verified agreement. IWC runtime and reasoning come from invocation records; its all-zero replicate sets describe this detector under different recorded helper lists, not a stable agent preference. Archive table X13.

Supplementary Table 26 | Why choose a user-defined tool? Distinguishing capability gaps from interface friction. Purposive mechanism cases, not percentages of all custom-tool use. `results.deep.case_sources` retains exact trace paths, hashes and line-numbered agent statements. The treeness counterexample establishes scored correct output with retained `PhyKIT` jobs and no detected named UDT request, not an exhaustive native-only execution chain. To distinguish necessity from choice prospectively, snapshot catalog/search results, record scored incorrect alternatives and parameter errors, then compare repeated matched tasks with a fixed tool catalog and an independently validated native workflow. The current archive cannot estimate the fraction used because no suitable Galaxy tool existed. Archive table X15.

Supplementary Table 27 | Observed helper exposure and Tool Shed use in IWC. No named `run_galaxy_udt_and_wait` event or request-linked UDT job was detected. That helper is absent from all 120 recorded `galaxy_mcp_tools` lists. A helper-list omission is evidence about the recorded interface, not proof that every API or shell route to custom code was unavailable or unused. The Tool Shed classifier excludes the declared `UTILITY_REPOS` text/table codebook; it is not an independent audit of domain-tool coverage or task completion. Each of 118 detailed histories contains at least one qualifying successful job, which does not demonstrate that all required operations used such jobs. Two pseudobulk histories include Jupyter interactive-tool jobs. Local shell analysis, arbitrary code within utilities, and inherited jobs are not excluded. Native-tool sufficiency, agent preference and strict Galaxy-only execution remain unestablished; two metadata-only histories are excluded from job denominators (Supplementary Table 11). Archive table I11.

Supplementary Table 28 | Reliability of user-defined tools. Galaxy-condition runs with Codex execution traces (1,908 runs), which return a structured status for every tool-run call. Failure phase and error-message availability come from the failure record of each Galaxy job error. Probe tools are user-defined tools whose identifier names a probe, preflight, render, smoke, diagnostic or sanity check.

Supplementary Table 29 | Galaxy execution, missing evidence and candidate recoveries. Jobs are deduplicated by server/native job ID and exclude data-fetch jobs, but include preparation and legacy upload. Histories may contain inherited/later state. A candidate is later same-tool/input success, not verified recovery of a scientific objective. 31/32 collection-limited runs had runs scored correct. The original fresh-history check is false for all 750 Galaxy runs, despite retrieved history evidence; Galaxy-only completion remains unresolved. Archive table B4.

Supplementary Table 30 | Galaxy execution and recovery evidence. The archive contains 22,067 distinct creating jobs: 5,381 data-fetch and 16,686 Galaxy analysis (including 372 upload1 jobs). Galaxy analysis jobs are not independent scientific attempts. Nonzero shell exits occur in 676/1,188 available Galaxy transcripts and 882/1,200 open-ended code transcripts; probes/searches can return nonzero, so these are not comparative scientific-failure rates. The two metadata-only histories contain 196 error-state datasets in `state_ids` but zero in `state_details`, plus 17 unaccounted elements. Those dataset states are not added to creating-job totals. Archive table C3.

Supplementary Table 31 | Galaxy execution, missing evidence and candidate recoveries. Harmonized with Supplementary Table 29/Supplementary Table 30: jobs are deduplicated by server and native job ID, data fetch is excluded and legacy upload is kept. That gives 1,352 jobs, the IWC report's 1,330 non-acquisition jobs plus 22 upload1 jobs; error and deleted counts are identical. The two metadata-only histories are amplicon-denoising runs capped at 300 contents by the collector. Their failure counts are unknown, not zero. The 41 candidates collapse to 27 distinct successful endpoints and are not verified recoveries. No equivalent detector exists for open-ended code. Archive table I7.

Supplementary Table 32 | Specific recurring diagnostics and the mechanisms they suggest. Counts are deduplicated creating jobs; tool versions are pooled within the named family. The denominator contains every error job for that family, including missing/truncated diagnostics. Exact regex rules and matching job/source-event IDs are retained in the analysis JSON. These messages support input, datatype, identifier and design-validation hypotheses; they do not alone assign blame to the model, wrapper or platform. Multiple signatures may overlap. No claim is made that a tool always fails: the diagnostic probe with 57/57 error states in Supplementary Table 64 is not a biomedical-analysis success endpoint. Archive table X18.

Supplementary Table 33 | Which operational error messages are observable? All benchmark-specific paired model configurations, not just shared GPT labels. Deduplicated Galaxy job errors; case-insensitive regex matches on retained `stderr/stdout` excerpts, with overlapping categories. The exact codebook and every source event reference are in the analysis JSON. These are message indicators, not independently adjudicated root causes; missing/truncated logs suppress detection, and absent text does not mean no diagnostic existed on the server. No significance test compares these unequally observed taxonomies. Exit code zero does not override the recorded Galaxy error

state. In IWC, fastp and Bowtie2 Galaxy job errors retain no text at all (Supplementary Table 32), so the generic codebook under-detects its failures. Archive table X5.

Supplementary Table 34 | Causes of every failed Galaxy interface call (A1–A8: no job was created; B1–B5: a job was created and failed). All Galaxy-condition runs with execution traces, both agent harnesses. Rules are applied in the listed order to the error text and status of each failed call; the first match assigns the cause. Rates use all Galaxy interface calls in the benchmark as denominator. Example strings are verbatim, truncated and screened for credentials.

Supplementary Table 35 | Execution condition comparison using a shared, limited operational marker. Each pair requires six transcripts with at least one numeric shell exit code per run. The superseded Claude Code model configuration is excluded because its exit codes are unobserved. Intervals resample capsules/tasks with all eligible model configuration bundles. Only the shell channel is compared: searches, probes and package checks can return nonzero, while Galaxy job failures can be returned as successful API calls. Thus this marker does not answer whether Galaxy has more scientific failures or is intrinsically harder. Higher Galaxy input-token use (Supplementary Table 49/Supplementary Table 50), server-job errors (Supplementary Table 29/Supplementary Table 30) and concrete wrapper-binding friction (Supplementary Table 26) are distinct, supported forms of burden. Archive table X10.

Supplementary Table 36 | Adjudicated examples: correction versus changed objective. Purposive positive and contradictory examples, not a random sample of 213 candidates. Exact event IDs, parameter changes, timestamps and excerpts are preserved in case_reviews in the CompBioBench audit. No recovery success percentage is inferred from these two cases. Archive table C7.

Supplementary Table 37 | Parameter substitution in tool-run calls. The Galaxy interface compares the parameter values an agent requested with the tool state Galaxy resolved from them. A parameter substitution is either blocked before submission by the benchmark's check (status validation_parameter_mismatch) or executed (status parameter_mismatch). "Value replaced" means a requested value was changed; "requested value dropped" means a requested parameter is absent from the resolved state. Codex execution traces only. Short tool identifiers are Tool Shed repository names.

Supplementary Table 38 | Galaxy improvements suggested by the observed evidence. Priorities are analyst proposals, not measured intervention effects or claims that these features are wholly absent. Galaxy already documents history API and exports (https://docs.galaxyproject.org/en/release_26.1/_modules/galaxy/webapps/galaxy/api/histories.html), error troubleshooting (https://training.galaxyproject.org/training-material/faqs/galaxy/analysis_troubleshooting.html), datatype handling (https://training.galaxyproject.org/training-material/faqs/galaxy/datatypes_understanding_datatypes.html) and workflow reports (https://training.galaxyproject.org/training-material/faqs/galaxy/workflows_report_view.html). The proposed work is to assess and extend these capabilities at the agent interface. Collection limits belong to this audit's collector and are not established Galaxy server limits. Documentation checked 2026-09-23; no deployed-server feature audit was performed. Archive table X7.

Supplementary Table 39 | Common software families: installed Galaxy wrappers versus code-command indicators. Three shared GPT model configurations. The last six families were added for IWC workflows; earlier rows are unchanged. Galaxy counts runs whose retained jobs name a Tool Shed wrapper from the family; open-ended code uses a declared command-name codebook, including PyDESeq2 as a DESeq2-family implementation. This inventory codebook is separate from the unchanged path-fingerprint vocabulary; shared family labels do not imply equivalent implementations or versions. Denominators require detailed histories or retrieved transcripts, respectively. Indicators are nonexclusive and their visibility differs: libraries inside custom tools can be hidden, and a command mention is not a verified invocation. This answers which families are observable in both conditions; it is not a fair count of equivalent scientific operations or proof of absent software. Workflow-derived task origin motivates checking tool coverage, but does not establish that every required tool/version was available on the recorded server. Archive table X9.

Supplementary Table 40 | Declared solution routes versus output agreement. Method declarations were extracted on four tasks; 95/96 are available. Route counts use complete three-declaration replicate sets, whereas means use all numeric runs in each row, including the incompletely declared replicate set. Host-removal means have unequal model composition. Routes are normalized declarations (for example Bowtie2 plus MACS2, DESeq2 or edgeR), not verified executions. Different declared routes coexist with near-ceiling output agreement, which supports scoring valid alternatives against route-specific references. Route-specific references also mean output agreement is conditional on route registration (Supplementary Table 13). Archive table I12.

Supplementary Table 41 | Observable tool-set agreement within replicate sets under the harmonized rule. One replicate set = task x model configuration x execution condition with three replicate labels. This report harmonizes BOTH benchmarks: all three fingerprints must be observed and nonempty. Identical / two identical / all distinct partition evaluable replicate sets. Jaccard = intersection/union, averaged over the three replicate pairs and then over eligible replicate sets. Galaxy uses version-stripped job tool IDs (excluding data fetch, retaining legacy upload); code uses the archived closed command vocabulary. These instruments cannot rank scientific consistency across execution conditions; order, most parameters and biological validity are unmeasured. BixBench counts differ from its legacy report because that report also admitted partly missing/empty fingerprints. Archive table B5.

Supplementary Table 42 | Observable tool-set agreement within replicate sets. One replicate set = task x model configuration x execution condition with three replicate labels. This report harmonizes BOTH benchmarks: all three fingerprints must be observed and nonempty. Identical / two identical / all distinct partition evaluable replicate sets. Jaccard = intersection/union, averaged over the three replicate pairs and then over eligible replicate sets. Galaxy uses version-stripped job tool IDs (excluding data fetch, retaining legacy upload); code uses the archived closed command vocabulary. These instruments cannot rank scientific consistency across execution conditions; order, most parameters and biological validity are unmeasured. BixBench counts differ from its legacy report because that report also admitted partly missing/empty fingerprints. Archive table C5.

Supplementary Table 43 | Observable tool-set agreement within replicate sets under the harmonized rule. One replicate set = task x model configuration x execution condition with three replicate labels. This report harmonizes all three benchmarks: all three fingerprints must be observed and nonempty. Identical / two identical / all distinct partition evaluable replicate sets. Jaccard = intersection/union, averaged over the three replicate pairs and then over eligible replicate sets. Galaxy uses version-stripped job tool IDs (excluding data fetch, retaining legacy upload); code uses the archived closed command vocabulary. These instruments cannot rank scientific consistency across execution conditions; order, most parameters and biological validity are unmeasured. The legacy IWC fingerprint output is not used. Declared biological routes are in Supplementary Table 40. Archive table I8.

Supplementary Table 44 | Split replicate sets by benchmark, execution condition and model configuration. A replicate set is the replicate runs of one task × model configuration × execution condition; it is split when its replicate runs disagree in outcome (definitions in a). CompBioBench uses the 82 strong-consensus tasks. The superseded Claude Code model configuration was run on BixBench-Verified-50 only.

Supplementary Table 45 | Task specification size and recorded workload versus operational failure markers. Three shared GPT model configurations; nine observed runs are required per task/execution condition. Prompt word count is a pre-execution task-description-size proxy, not a validated difficulty score. Median job count is a post-execution workload measure; retries increase it, creating reverse causation. The job response divides recorded errors by recorded jobs to reduce the automatic opportunity effect of simply counting failures. Intervals use 5,000 capsule/task bootstrap resamples, seed 20260922, with ties reranked within each resample. These exploratory, unadjusted associations cannot establish that task complexity causes errors. Independent difficulty labels, budgets and attempt-level chronology are needed. Archive table X11.

Supplementary Table 46 | Different tool-set fingerprints, identical submitted answers: does the observation hold? All benchmark-specific paired model configurations; three observed nonempty fingerprints are required. Answer identity uses exact archived submitted text after outer whitespace removal; it does not merge numerically close strings. Path variation means tool-ID/command-token sets differ, not independently adjudicated biological algorithms. BixBench can confirm accuracy; CompBioBench cannot confirm validity. The counts support multiple observed execution routes to the same answer, with these measurement limits. IWC is excluded: its final chat message is not the scored artifact; declared routes versus continuous output agreement are in Supplementary Table 40. Archive table X16.

Supplementary Table 47 | Does a consistent tool-set fingerprint imply accuracy? Descriptive association conditional on observable nonempty paths. Replicate sets share tasks; failure-prone task composition can drive the pattern. Agreement is neither correctness nor causal evidence that varied methods help. The retained rows allow inspection of both scored correct and scored incorrect consistent paths. Archive table B6.

Supplementary Table 48 | Divergence mechanism of every scored-incorrect replicate run in a split BixBench-Verified-50 replicate set. Every scored-incorrect replicate run in a split BixBench-Verified-50 replicate set (45 open-ended code condition runs in 33 sets; 36 Galaxy-condition runs in 28 sets) was compared call by call with the scored-correct replicate run(s) of the same set. The divergence mechanism names what differed between the replicate runs, not why the answer was incorrect (see the primary cause in the failure ledger). Definitions in Supplementary Note 8.

Supplementary Table 49 | Input-token usage by model configuration. Ratio = median input tokens of three Galaxy runs / median of three open-ended code runs for the same task and model configuration; all six totals must be available. Q1-Q3 are quartile endpoints. Absolute medians use every available run in that row and are not the numerator/denominator of the paired median ratio. Cached input is already included. Output/reasoning are not added. Usage covers archived primary turns, not full campaign, compute or monetary cost. Failed runs are retained; missing usage is never zero. Archive table B7.

Supplementary Table 50 | Input-token usage by model configuration. Ratio = median input tokens of three Galaxy runs / median of three open-ended code runs for the same task and model configuration; all six totals must be available. Q1-Q3 are quartile endpoints. Absolute medians use every available run in that row and are not the numerator/denominator of the paired median ratio. Cached input is already included. Output/reasoning are not added. Usage covers archived primary turns, not full campaign, compute or monetary cost. Failed runs are retained; missing usage is never zero. Fourteen of 400 pairs are excluded (3 GPT-5.5, 10 Luna, 1 DeepSeek). The runtime-and-reasoning-verified sensitivity contains only 99 DeepSeek pairs (median 3.53), so it changes the model configuration population. Astra usage is available in 100/100 unpaired open-ended code runs (median input 314,244.5); no Galaxy ratio is possible. Archive table C6.

Supplementary Table 51 | Input-token usage by model configuration. Ratio = median input tokens of three Galaxy runs / median of three open-ended code runs for the same task and model configuration; all six totals must be available. Q1-Q3 are quartile endpoints. Absolute medians use every available run in that row and are not the numerator/denominator of the paired median ratio. Cached input is already included. Output/reasoning are not added. Usage covers archived primary turns, not full campaign, compute or monetary cost. Failed runs are retained; missing usage is never zero. IWC usage is the audited terminal record, one per run, 240/240. Typical usage is higher in Galaxy, but summed input is lower: 815.8 million in Galaxy versus 919.6 million in open-ended code, because of a long open-ended code upper tail. The ratio medians reproduce the IWC report's 1.66-2.77 within-task ratios. With ten task clusters, every IWC interval includes one, so the typical direction is uncertain across the resampled task population. These intervals do not estimate monetary or causal overhead. Archive table I9.

Supplementary Table 52 | Does relative input-token overhead generalize across benchmarks? The contrast columns divide two benchmark-specific medians of within-task ratios; they are not absolute-token ratios. A value above one means greater relative overhead than in BixBench. IWC has ten task clusters, so its intervals are wide. Its lower point estimate cannot be attributed to installed wrappers: prompts, budgets, task selection and open-ended code tails also differ (Supplementary Table 51). Benchmark strata are resampled separately, preserving tasks/model configurations within clusters. Selection differs and these are observational comparisons. Intervals as in Supplementary Note 9. Archive table X2.

Supplementary Table 53 | Tool search and inspection burden, and direct Galaxy API calls. Galaxy-condition runs with Codex execution traces. Tool search and inspection are search_galaxy_tools and inspect_galaxy_tool calls. Returned text is the number of characters returned to the agent, used as a proxy for context consumed because input-token usage cannot be attributed to individual calls. Inspected-but-never-executed counts distinct tool identifiers per run that were inspected but not submitted, summed over runs. A shell command is a direct Galaxy API call when it uses the BioBlend library or calls the Galaxy histories, datasets, jobs or tools API; operations are matched among commands referencing BioBlend or any /api/ path. The provided analysis history exists only in BixBench-Verified-50.

Supplementary Table 54 | Model difference: accuracy difference and token use relative to GPT-5.5. Reference is GPT-5.5 within the same execution condition and the same 50 tasks. Accuracy differences average all three replicate runs; token ratios divide task medians of three replicate runs and are then summarized by the median ratio. Error-run differences use the subset of tasks with all six detailed Galaxy runs and compare proportions with at least one error; the error column can therefore have a different task population from the other columns. Intervals use 20,000 paired capsule bootstrap resamples, seed 20260922. They are exploratory and pointwise. A difference interval containing zero does not establish equal performance or non-inferiority. Within Galaxy, Sol and Luna consumed more input tokens than GPT-5.5 without a resolved accuracy advantage; both DeepSeek model configurations consumed more and had lower observed accuracy. These are selected-corpus associations, not general model rankings. Absolute pooled medians in Supplementary Table 58 and median paired

ratios here can order models differently because they are different estimands. Archive table B15.

Supplementary Table 55 | All-task output agreement, replicate dispersion, operational markers and tokens. IWC counterpart to Supplementary Table 58 and Supplementary Table 61, using continuous output agreement instead of accuracy. Luna ran at max reasoning and the others at high, so model configuration and reasoning are confounded. Budgets were unbalanced: 6 h or 12 h ceilings differ by model configuration and execution condition. Galaxy job errors and nonzero shell exits are different instruments, as in Supplementary Table 58. These are all-ten-task available-case summaries, not the matched population in Supplementary Table 7: GPT-5.5 code has nine scored tasks, other rows ten. Do not subtract unequal-population means. These are selected-corpus associations, not general model rankings or measured monetary costs. Archive table I13.

Supplementary Table 56 | Repeatability of runs scored correct. Each row contains 50 task replicate sets; the middle three categories are mutually exclusive. At least one is an overlapping endpoint, not single-run accuracy. These are replicate-labelled archived runs, not best-of-three adaptive attempts. Archive table B2.

Supplementary Table 57 | Largest accuracy discordances and tasks scored incorrect throughout. Selection rule: absolute accuracy difference $\geq 5/15$ or zero accuracy in both execution conditions; all five qualifying tasks are shown. Task-linked jobs are deduplicated within task. Operationally successful jobs do not establish a correct answer: RCV and transition/transversion results were scored incorrect despite no recorded job errors. RCV means relative composition variability. The source reports implicate threshold choices, upstream gene sets and RCV input construction; those mechanisms are not independently adjudicated here. Archive table B3.

Supplementary Table 58 | Model configuration outcomes, repeatability, operational markers and tokens. Each task contributes three runs/model configuration/execution condition. Scored correct-answer reliability is used for BixBench; CompBioBench can only report answer-text consistency, never correctness. Galaxy job errors and nonzero shell exits are different instruments. A shell exit may be a probe or failed search; Galaxy analysis is often offloaded to server jobs, so fewer shell exits do not establish easier execution. Shell denominators require at least one numeric recorded exit code: the superseded Claude Code traces contain no numeric exit codes and are unavailable, not zero failures. Tokens describe archived primary turns, not full execution cost. Missing data are excluded with explicit denominators; all observed outcomes are included. Archive table B14.

Supplementary Table 59 | Model configuration-specific exclusive task successes hidden by pooling models. Same rule as Supplementary Table 4, but now applied within a task and model configuration: one or more scored correct replicate runs versus none. All 13 qualifying pairs are shown: nine Galaxy-exclusive pairs and four open-ended code-exclusive pairs out of 250. These do not contradict Supplementary Table 4 because another model configuration can succeed in the other execution condition. A one-of-three result is weak repeatability, not a stable platform advantage. Replicate sets share tasks and cannot be treated as independent observations in a simple binomial test. Rows without an audited mechanism remain unresolved rather than being attributed to missing tools or model knowledge. Archive table B16.

Supplementary Table 60 | Usage completeness and repeated-answer consistency. Usage denominators are 300 runs/row; answer counts partition 100 replicate set task replicate sets/row. Answers use archived submitted text with outer whitespace removed, not raw-file hashes. Text identity does not establish semantic equivalence or correctness. Among all 12 answers/task/execution condition, 62/100 Galaxy tasks and 58/100 open-ended code tasks have one distinct answer; answer sets differ across execution conditions in 43/100 tasks. Archive table C2.

Supplementary Table 61 | Model configuration outcomes, repeatability, operational markers and tokens. Each task contributes three runs/model configuration/execution condition. Scored correct-answer reliability is used for BixBench; CompBioBench can only report answer-text consistency, never correctness. Galaxy job errors and nonzero shell exits are different instruments. A shell exit may be a probe or failed search; Galaxy analysis is often offloaded to server jobs, so fewer shell exits do not establish easier execution. Shell denominators require at least one numeric recorded exit code: the superseded Claude Code traces contain no numeric exit codes and are unavailable, not zero failures. Tokens describe archived primary turns, not full execution cost. Missing data are excluded with explicit denominators; all observed outcomes are included. Archive table C9.

Supplementary Table 62 | Outlier case: what can actually be said about biomedical knowledge? The task asks which of ten tagAlign files comes from a different assay. Missing item scores prevent saying that most agents were wrong or assigning a biomedical-knowledge error rate. In GPT-5.5 Galaxy replicate 1, the trace records an unsuccessful external checksum lookup, then binned coverage correlations, chromosome distribution, strand correlation and transcription-start-site enrichment; it explicitly considers replicate set-line copy-number differences as an alternative explanation. These are documented diagnostic choices, not proof that the final index 9 is correct. Source: task evidence; trace lines 29, 30, 38 and 44 in results.deep.case_sources. Operational errors and uncertain scientific interpretation must remain separate. Archive table C10.

Supplementary Table 63 | Run-level operational burden with the shared model configuration mix. Three shared GPT labels only; missing detailed histories are excluded explicitly. Each run contributes one binary indicator, preventing runs with many jobs from dominating this estimate. Histories may contain inherited/latent state; this is recorded burden, not a causal benchmark/platform failure probability. No code-condition counterpart uses an equivalent job instrument. Archive table X4.

Supplementary Table 64 | Where recorded Galaxy job errors concentrate. Top five tool IDs ranked by error-job count within each benchmark (ties ordered by full ID); all Galaxy analysis jobs and all model configurations. These are triage targets, not tool-quality rankings: exposure, parameters, input quality, custom wrappers and task mix differ. In particular, udt-render-probe-v1 identifies a diagnostic probe; its error states cannot be assumed to be failed biomedical analyses. IWC error concentrations sit in domain workflow wrappers, the tools its tasks were built around. Full tool IDs and creating-job references are retained in jobs in the analysis JSON. Archive table X6.

Supplementary Table 65 | Claim strength, remaining questions and evidence needed. All 160 task evidence paths, hashes, included run IDs and original finding IDs are linked in the source manifest. Table-specific display values and calculated statistics, run/replicate set denominators, job references and exclusion rules are in the analysis JSON. Source task evidence is unchanged; no hidden references, recovered agent code, new Galaxy runs or new correctness judgements were used. Archive table X8.

Supplementary Table 66 | Testing the six motivating observations. These verdicts evaluate the motivating statements, not a prespecified hypothesis set. All statistical intervals are exploratory. Model capacity, workbench functionality, operational friction, scientific correctness and execution cost remain separate endpoints. Archive table X17.

Supplementary Table 67 | Distinct benchmark endpoints in the shared-label descriptive set. Three shared supplied GPT labels, not verified identical runtime/reasoning settings across benchmarks. Endpoints differ, so read rows side by side, never pooled or ranked. BixBench differences are percentage points of accuracy; IWC differences are output agreement units on 0-1. All IWC columns use the same nine-task subset as Supplementary Table 7; host-removal nulls and conflicts are documented in Supplementary Table 13, not relabelled as failures. A BixBench zero denotes evaluator rejection; an IWC zero can reflect a contract check or output disagreement and is not an interchangeable scientific-failure endpoint. CompBioBench answer-text identity is omitted from this outcome table because it is neither a ceiling score nor correctness (see Supplementary Table 60). Task sets, prompts, budgets, interfaces and endpoints differ; whether workflow-derived task origin narrows an execution condition effect is not identifiable from this comparison. Archive table X20.

Supplementary Data legends

Supplementary Data 1 | Per-run summary of all 4,240 runs. One row per run with benchmark, task, model configuration, execution condition, replicate run, evaluator verdict or output agreement, submitted answer, verifier mode, input-token usage, shell and Galaxy interface call counts and failed-call counts by cause. Reference answers are omitted to limit benchmark contamination.

Supplementary Data 2 | All 7,389 failed Galaxy interface calls. One row per failed call with run, trace line, interface tool, status, cause, verbatim error text and arguments (truncated to 1,500 characters; screened for credentials).

Supplementary Data 3 | 2,085 parameter substitutions with a replaced value. For each tool-run call with a Codex execution trace in which a requested value was replaced, the first replaced parameter with the value requested and the value Galaxy resolved.